

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I E.Ragul (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Ragul

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- D1 cannot work Night shift
- D2 must work before D3 (Morning < Afternoon < Night)
- Only one doctor per shift
- D3 cannot work Morning
- All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Answer:

Backtracking Search Solution:

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints

D1 cannot work **Night**.

D2 must work **before** D3 (Morning < Afternoon < Night).

Only **one doctor per shift**.

D3 cannot work **Morning**.

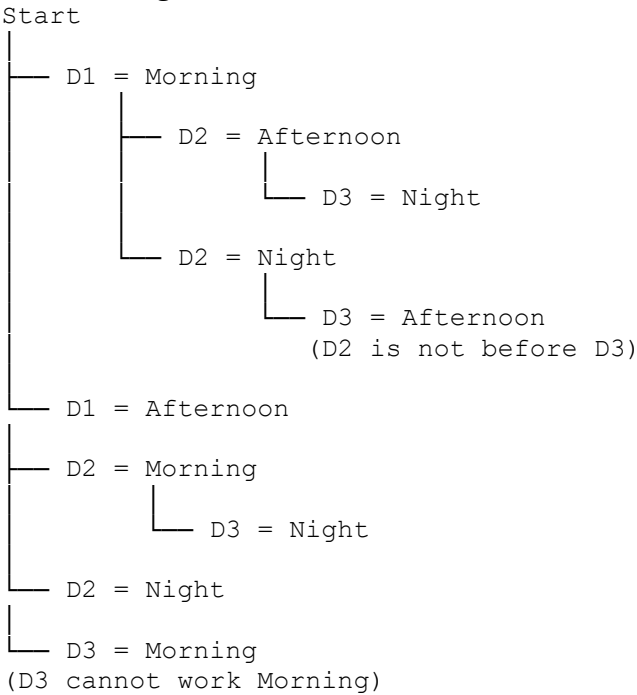
Every doctor must be assigned **exactly one shift**.

Step	Doctor Assigned	Possible Shifts	Chosen Shift	Current Assignment	Remaining Shifts	Constraint Check
Step 1	D1	Morning, Afternoon (Night not allowed)	Morning	D1 → Morning	Afternoon, Night	D1 cannot work Night ✓
Step 2	D2	Afternoon, Night	Afternoon	D1 → Morning D2 → Afternoon	Night	Shift available ✓
Step 3	D3	Night (only remaining shift)	Night	D1 → Morning D2 → Afternoon D3 → Night	None	D3 not Morning ✓ D2 before D3 ✓ One doctor per shift ✓ All doctors assigned ✓

Constraint Checking Table:

Step	Assignment	Constraint Check
1	D1 = Morning	D1 not Night
2	D2 = Afternoon	Shift available
3	D3 = Night	D3 not Morning
3	D2 before D3	Afternoon < Night
3	One doctor per shift	All shifts unique
3	All doctors assigned	Each assigned once

Backtracking Search Tree



Final Valid Schedule:

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Morning → D1

Afternoon → D2

Night → D3

This is a valid assignment found using **Backtracking Search**, satisfying all the given constraints.
A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles
- Use Manhattan Distance heuristic

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Answers:

BFS Search Algorithm

Grid Environment (5 × 5)

```
S - - X -
- X - X -
- X - - -
- - X X -
- - - G -
```

Legend:

S = Start (0,0)

G = Goal (4,3)

X = Obstacle

- = Free cell

Constraints:

Movement allowed: Up, Down, Left, Right

Cost of each move = 1

Cannot pass through obstacles

Manhattan Distance Heuristic:

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

Step-by-Step BFS Node Expansion:

Step	Expanded Node	New Nodes Added to Queue	Queue After Update
1	(0,0)	(0,1), (1,0)	[(0,1), (1,0)]
2	(0,1)	(0,2)	[(1,0), (0,2)]
3	(1,0)	(2,0)	[(0,2), (2,0)]
4	(0,2)	(1,2)	[(2,0), (1,2)]
5	(2,0)	(3,0)	[(1,2), (3,0)]
6	(1,2)	(2,2)	[(3,0), (2,2)]
7	(3,0)	(3,1), (4,0)	[(2,2), (3,1), (4,0)]
8	(2,2)	(2,3)	[(3,1), (4,0), (2,3)]
9	(3,1)	(4,1)	[(4,0), (2,3), (4,1)]
10	(4,0)	No new node	[(2,3), (4,1)]
11	(2,3)	(2,4), (3,4)	[(4,1), (2,4), (3,4)]
12	(4,1)	(4,2)	[(2,4), (3,4), (4,2)]

Step	Expanded Node	New Nodes Added to Queue	Queue After Update
13	(4,2)	Goal (4,3)	[(2,4), (3,4), (4,3)]

Optimal Path:

Step	Path
1	(0,0) → (1,0)
2	(1,0) → (2,0)
3	(2,0) → (3,0)
4	(3,0) → (3,1)
5	(3,1) → (4,1)
6	(4,1) → (4,2)
7	(4,2) → (4,3) (Goal)

Path Cost:

Each move costs 1.
 Total moves = 7
 Total Cost = 7

Final Answer:

Algorithm Used: Breadth-First Search (BFS)
 Optimal Path:
 (0,0) → (1,0) → (2,0) → (3,0) → (3,1) → (4,1) → (4,2) → (4,3)
 Total Path Cost: 7
 Result: BFS successfully finds the shortest path while avoiding obstacles.

Constraints:

The robot can move up, down, left, right
 Some cells are blocked (obstacles) and cannot be traversed
 The robot has limited sensing capability (can only observe adjacent cells)
 Each movement has a cost of 1, but entering risky zones has an additional cost of 2
 The robot must minimize total cost while ensuring safe navigation
 The robot must update its knowledge dynamically (online search scenario)

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Answer:

a) Analysis of Constraints and Their Effect on Decision-Making

Constraint	Effect on Robot's Decision-Making
Robot can move Up, Down, Left, Right	The robot considers only four possible actions at each step.
Obstacles cannot be traversed	The robot avoids blocked cells and searches for an alternative path.
Limited sensing (adjacent cells only)	The robot cannot see the entire environment and must update its map while moving.
Movement cost = 1	Every normal move increases the total path cost by 1.
Risky zones have additional cost of 2	The robot prefers safer paths unless the risky path gives a lower overall cost.
Minimize total cost	The robot selects the least-cost path instead of simply the shortest path.
Dynamic knowledge (Online Search)	The robot continuously updates its decisions whenever it discovers new obstacles or risky areas.

b) Solution Approach Using Uniform Cost Search (UCS) with Online Search

Step 1: Start

Place the robot at the start position S.

Initialize the total cost as 0.

Step 2: Sense the Environment

Observe only the adjacent cells (Up, Down, Left, Right).

Identify free cells, obstacles, and risky zones.

Step 3: Expand Possible Moves

Generate all valid neighboring cells.

Ignore blocked cells.

Step 4: Calculate Path Cost

Normal cell = Cost **1**

Risky cell = Cost **3** (1 movement + 2 risk)

Step 5: Choose Minimum-Cost Path

Use **Uniform Cost Search** to expand the node with the lowest accumulated cost.

Step 6: Update Knowledge

If a new obstacle or risky zone is detected, update the map.

Recalculate the best path from the current position.

Step 7: Repeat

Continue sensing, updating, and moving until the robot reaches the survivor **G**.

Flow of the Algorithm:

Start



Sense Adjacent Cells



Generate Valid Moves



Calculate Path Cost



Select Lowest-Cost Node (UCS)



Move to Next Cell



Update Environment Information



Goal Reached?



Yes → Rescue Survivor

No → Repeat

c) AI Concepts Applied

1. Intelligent Agent

The rescue robot acts as an **Intelligent Agent**.

It senses the environment using sensors.

It makes decisions based on available information.

It performs actions (movement) to achieve the goal.

2. Rationality

The robot is **rational** because it:

Chooses the least-cost path.

Avoids obstacles.

Minimizes travel through risky areas.

Updates its decisions when new information becomes available.

3. Environment Type

Property	Description
Partially Observable	The robot can observe only adjacent cells.
Dynamic	The environment knowledge changes as the robot explores.
Sequential	Each action affects future decisions.
Discrete	The environment is represented as grid cells.
Single Agent	Only one rescue robot performs the task.

d) Justification of the Chosen Approach

Uniform Cost Search (UCS) combined with an **Online Search Agent** is the most suitable approach because:

It always finds the **minimum-cost path**.

It considers both movement cost and additional risk cost.

It can adapt to newly discovered obstacles and risky zones.

It works effectively in **partially observable environments**.

It guarantees an optimal solution when all costs are positive.

Therefore, UCS with Online Search enables the rescue robot to safely and efficiently reach the survivor while minimizing the total navigation cost.

Final Answer:

Search Strategy: Uniform Cost Search (UCS) with Online Search Agent

Agent Type: Intelligent Agent

Environment: Partially Observable, Dynamic, Sequential, Discrete, Single-Agent

Objective: Reach the survivor while minimizing total movement and risk cost.

Reason for Choosing UCS: It guarantees the least-cost path and adapts dynamically to newly discovered information, making it ideal for disaster rescue scenario

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided